2

ELASTICITY OF DEMAND

QUICK REVISION

Elasticity of Demand: Elasticity of demand measures the responsiveness of the quantity demanded of a commodity of change in its price, change in consumers' income and price of other related commodities. Accordingly, there are three types of elasticity of demand. These are:

(A) Price elasticity of demand,
(B) Income elasticity of demand, and
(C) Cross elasticity of demand.

(A) Price Elasticity of Demand: The responsiveness of quantity demanded of a good or service to change in its price is known as price elasticity of demand (ep).

Different Price Elasticities of Demand:

Table

S.No.	Degrees of price elasticity of demand	Type	Response in Quantity Demanded	Shape of the Demand Curve
1.	$e_p > 1$	Relatively elastic	Percentage change in demand is greater than percentage change in price.	Relatively flat
2.	$e_p < 1$	Relatively inelastic	Percentage change in demand is less than percentage change in price.	Relatively steep
3.	$e_p = 1$	Unit elastic	Percentage change in demand is equal to percentage change in price.	Rectangular hyperbola
4.	$e_p = 0$	Perfectly inelastic	No change in demand at all in response to change in the price.	Vertical
5.	$e_p = \infty$	Perfectly elastic	Demand is infinite at the prevailing price.	Horizontal

1. Comparison of Elasticity: When two demand curves intersect each other the elasticity associated with flatter is greater.

2. Factors Affecting Elasticity of Demand:

(i) Nature of the commodity
(ii) Availability of substitutes
(iii) Number of uses
(iv) Possibility of postponement
(v) Proportion of income spent
(vi) Habits
(vii) Time period
(viii) Price level

3. Importance of Elasticity of Demand:

(i) Useful to a producer
(ii) Importance in factor pricing
(iii) Useful to a finance minister
(iv) Importance in international trade
(v) It explains paradox of poverty

4. Methods of Measurement:

(i) Percentage Method: $ep = \frac{\text{Percentage change in demand}}{\text{Percentage change in price}}$

(ii) Total Outlay Method: According to this method:
When price and total outlay are inversely related, ep will be greater than one.
When price and total outlay are positively related, ep will be less than one.
When total outlay remains unchanged when price of the commodity changes, ep will be equal to one.

(iii) Geometrical Method: $ep = \frac{\text{Lower Line Segment from the point}}{\text{Upper line segment from the point}}$

(a) At mid-point of demand curve, ep = 1
(b) At any point above the mid-point ep > 1
(c) At any point below the mid-point ep < 1
(d) At a point where demand curve touches the Y-axis, ep = ∞
(e) At a point where demand curve touches X-axis, ep < 1

NOTE

Percentage method gives exact measurement of price elasticity of demand while total outlay method gives us only rough estimates of price elasticity.

(B) Income Elasticity: Degree of responsiveness of quantity demanded of a commodity to the change in income of the consumer. In other words, income elasticity of demand measures how responsive the quantity demanded of a commodity is to a change in income, price of the goods remaining unchanged. In case of normal goods, the algebraic sign of this elasticity is positive and is negative in case of inferior goods.

Table

S.No.	Numerical measure	Description
1.	Negative	Quantity demanded decreases as income increases (e.g., inferior goods).
2.	Zero	Quantity demanded remains unchanged as income increases (e.g., inexpensive goods).
3.	Greater than zero but less than one	Quantity demanded rises less than in proportion to income increase.
4.	Unity	Quantity demanded rises in the same proportion as rise in income.
5.	Greater than one	Quantity demanded rise more than in proportion to income increase.

(C) Cross Elasticity : Degree of responsiveness of quantity demanded of commodity X to the change in the price of commodity Y. Cross elasticity measures how responsive demand of one good is to a change in the price of another goods.

1 MARK QUESTIONS

A. FILL IN THE BLANKS

1._____sign shows the inverse relationship between price and quantity demanded.
2. 6% change in price leads to an 18% change in quantity demanded. Price elasticity of demand is _____
3. Using geometric method, elasticity of demand = _____.
4. As we go down on a straight line demand curve, elasticity goes on _____.
5. Unitary elastic demand curve is _____ curve.
6. When change in price leave no impact on quantity demanded of a commodity, elasticity of demand is _____.
7. When the percentage change in price is less than percentage change in quantity demanded, the demand is _____.
8. When the percentage change in price is more than percentage change in quantity demanded, the demand is _____
9. A commodity having more substitutes _____ will be elastic.
10. Necessities are _____ elastic.
11. When price falls from ₹20 to ₹ 10 leading to 50% change in quantity demanded, elasticity of demand is _____.
12. Milk is _____ elastic due to _____ of milk.
13. Demand for salt is _____.
14. If elasticity of demand is, 10% fall in price will lead to _____ rise in quantity demanded.
15. If $e_d = 2$ and percentage change in price is 10 percentage. Percentage change in quantity demanded will be _____ .
16. When demand curve is parallel to Y axis, ED would be _____.
17. With a fall in price, total expenditure remains constant, ED would be _____.
18. When price and total expenditure move in the same direction then ED will be _____.
19. To calculate ED by point method on a non-linear demand curve, a _____ has to be drawn.
20. When Ed = 2 and price falls by 5% what will be change in demand _____.

ANSWERS

1. Minus	2. 3	3. Lower line segment/ upper line segment	4. Decreasing
5. Rectangular hyperbola	6. Zero	7. greater than one	8. Less than one
9. Highly	10. Inelastic or less elastic	11. One	12. highly elastic, several uses
13. Inelastic	14. 10 %	15. 20 %	16. Zero
17. One	18. Inelastic	19. Tangent	20. 10%

B. TRUE OR FALSE

1. When quantity demanded of a commodity does not change with change in price, then price elasticity of demand is zero.
Ans. **[True]** It happens in case of perfectly inelastic demand.

2. A commodity with large number of close substitutes shows high elasticity of demand.
Ans. **[True]** As with increase in price of the commodity having close substitutes consumer can easily switch over to a cheaper substitute.

3. Total expenditure on a commodity will always fall when price of a commodity falls.
Ans. **[False]** It is not necessary. When price falls total expenditure may increase also. to suga

4. When percentage change in demand is more than the percentage change in price then the demand curve is steeper.
Ans. **[False]** Demand curve will be flatter as price elasticity in this case is more than one.

5. In case of horizontal straight line demand curve, demand does not change with the change in price of the commodity.
Ans. **[False]** Because in case of horizontal straight line demand curve price remains constant and demand changes and with a little increase in the price demand becomes zero. Elasticity of demand in this case is infinite.

6. A flatter demand curve is more elastic than a steeper demand curve at the point of intersection.
Ans. **[True]** Flatter demand curve is more elastic because with same change in price, change in demand in case of flatter curve will be more in comparison to steeper curve.

7. On a downward sloping straight line demand curve, the elasticity of demand at the mid-point of the curve is unity.
Ans. **[True]** Elasticity of demand will be unity at mid-point of the demand curve as lower line segment is equal to the upper line segment.

8. In measuring price elasticity of demand, price is a dependent variable and quantity demanded is an independent variable.
Ans. **[False]** Price is independent variable and quantity demanded is a dependent variable.

9. The coefficient of price elasticity of demand is generally negative. However minus sign is ignored for the sake of convenience.
Ans. **[True]** As coefficient of price elasticity of demand is generally negative because of inverse relation between price and quantity demanded.

10. Commodity with several uses has highly elastic demand.
Ans. **[True]** As when the price increases the consumer can decrease the number of uses of that commodity and vice-versa.

11. On middle of a demand curve, geometric method explains unity elasticity of demand.
Ans. **[True]** Because at this point lower line segment is equal to the upper line segment.

12. If price falls by 10% and total expenditure rises by 10% then elasticity of demand will be one.
Ans. [False] If price falls and total expenditure rises then $E_D > 1$.

13. When demand curve is hyperbolic, Eo > 1.
Ans. **[False]** E0 = 1 at each point.

14. Eo for salt and newspaper is unity.
Ans. **[False]** For these goods E0 = 1.

15. Luxuries will have highly elastic demand.
Ans. **[True]** Because these demand can be postponed for future.

C. MULTIPLE CHOICE QUESTIONS

1. Relationship between price and demand in elasticity of demand is of ___ nature.
(a) qualitative (b) quantitative (c) competitive (d) none of these

2. Elasticity of demand means:
(a) % Δ in demand
(b) % Δ in income
(c) % Δ in price
(d) responsiveness of change in demand to a change in price

3. When demand curve is parallel to X-axis, what would be its elasticity of demand?
(a) E = α (b) E = 1 (c) E = > 1 (d) $E_D = 0$

4. When demand curve is parallel to Y-axis, E_D would be:
(a) $E_D = 1$ (b) E = 1 (c) E = > 1 (d) $E_D = 0$

5. Which of the following factor affects elasticity of demand?
(a) Nature of good (b) Level of income
(c) Availability of substitutes (d) All of these

6. With a change in price, when demand does not change, E0 will be:
(a) perfectly inelastic b) greater elastic ·(c) perfectly elastic (d) none of these

7. How do we symbolize greater elastic demand?
(a) $E_D > 1$ (b) $E_D < 1$ (c) $E_D = > \alpha$ (d) none of these

8. How do we symbolize less elastic demand?
(a) $E_D > 1$ (b) $E_D < 1$ (c) $E_D = 0$ (d) none of these

9. How do we write formula of measuring E_D?
(a) $\frac{Q}{P} x \frac{P}{Q}$ (b) $\frac{\Delta P}{\Delta Q} x \frac{Q}{P}$ (c) $\frac{\Delta Q}{\Delta P} x \frac{P}{Q}$ (d) none of these

10. What would be the nature of elasticity of demand for Giffen Goods?
(a) Positive (b) Negative (c) Neutral (d) None of these

11. How do we measure elasticity of demand with point method on a demand curve?

(a) $\frac{\text{Lower segment of demand curve from the point}}{\text{Upper segment of demand curve from the point}}$

(b) $\frac{\text{Upper segment of demand curve}}{\text{Lower segment of demand curve}}$

(c) $\frac{\text{Percentage change in demand curve}}{\text{Percentage change in price}}$

(d) None of these

12. What would be the nature of elasticity of demand for salt, newspaper and medicines?

(a) Elastic (b) Less Elastic (c) Greater elastic (d) Any of these

13. What would be the nature of elasticity of demand for luxury goods?

(a) Elastic (b) Less Elastic (c) Greater elastic (d) None of these

14. Which of the following has elastic nature of demand?

(a) Electricity (b) Fruits (c) Medicine (d) Pulses

15. When a demand is in downward sloping straight line, what would be elasticity of demand on middle of demand curve?

(a) $E_D = \alpha$ (b) $E_D = 1$ (c) $E_D = 0$ (d) $E_D => 1$

16. What would be elasticity of demand on a demand curve, at a point left to the middle point?

(a) $E_D = 1$ (b) $E_D => 1$ (c) $E_D < 1$ (d) $E_D => \infty$

17. When demand for sugar rises by 8% against a 5% fall in price, what would be its elasticity of demand?

(a) 1.5 (b) 0.62 (c) 1.6 (d) None of these

18. Which of the following formula is correct for measuring elasticity of demand?

(a) $\frac{Proportionate\ change\ in\ price}{Proportionate\ change\ in\ demand}$

(b) $\frac{Proportionate\ change\ in\ supply}{Proportionate\ change\ in\ supply}$

(c) $\frac{Proportionate\ change\ in\ demand}{Proportionate\ change\ in\ price}$

(d) None of these

19. When proportionate change in demand is greater with a small change in price, Ep would be:

(a) greater elastic (b) less elastic (c) unitary elastic (d) None of these

20. With a fall in price, when total expenditure increases, Ep will be:

(a) ED => 1 (b) ED <1 (c) ED=1 (d) ED=0

21. When total expenditure increases with a rise in price, E_D would be:

(a) ED => 1 (b) ED < 1 (c) ED=0 (d) ED=1

22. When price falls by 8% and total expenditure also rises by 8%, ED would be:

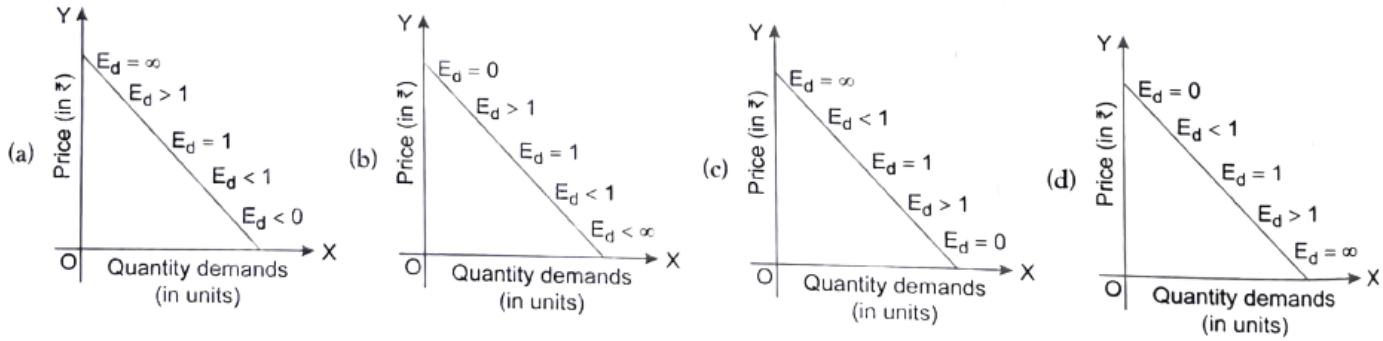
(a) ED = 1 (b) ED => 1 (c) Ep < 1 (d) None of these

23. With a fall in price, total expenditure remains constant, ED would be:
(a) $E_D=0$ (b) $E_D = 1$ (c) $E_D = \infty$ (d) ED > 1

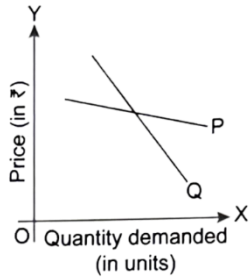
24. When Ep=2 and price falls by 5%, what would be increase in demand?
(a) 10% (b) 2.5 (c) 0.4 (d) None of these

25. ED=0.5, when price rises by 20%, what would be fall in demand?
(a) 10% (b) 40% (c) 5% (d) None of these

26. Which is the correct diagram according to the Geometric Method



27. If a commodity has large number of substitutes, then its demand curve will be:
(a) P
(b) Q
(c) P and Q both
(d) None of the above



28. Which of the following statements is correct?
(a) Price elasticity of demand is a pure number, as it is independent of the units of measurement.
(b) Price elasticity of demand is a negative number, as the demand for a good is negatively related of its price.
(c) Price elasticity of demand measures the magnitude of change in demand of a good as a result of change in its price.
(d) All of the above

29. Find the price elasticity in following example:

Price of X	Demand for X
5(P_1)	10(Q_1)
4(P_2)	15(Q_2)

(a)-2.5 (b)-2.0 (c) +3 (d) None of these

30. When a demand is in downward sloping straight line, what would be elasticity of demand on middle of demand curve?

(a) ED=a (b) Ep = 1 (c) ED=0 (d) ED => 1

31. A fall in the price of a commodity whose demand curve is a rectangular hyperbola causes total expenditure on the commodity to:

(a) Increase (b) Decrease (c) Remain unchanged (d) None of these

ANSWERS

1. (b)	2. (d)	3. (a)	4. (d)	5. (d)	6. (a)	7. (a)	8. (b)	9. (c)	10. (a)
11. (a)	12. (b)	13. (c)	14. (b)	15. (b)	16. (b)	17. (c)	18. (c)	19. (a)	20. (a)
21. (b)	22. (b)	23. (b)	24. (a)	25. (a)	26. (a)	27. (a)	28. (d)	29. (a)	30. (b)
31. (c)									

D. ASSERTION-REASON TYPE

Read the following statements-Assertion (A) and Reason (R), and select the correct alternative in each case:

(a) (A) is true, but (R) is false.
(b) (A) is false, but (R) is true.
(c) Both (A) and (R) are true and (R) is the correct explanation of (A).
(d) Both (A) and (R) are true but (R) is not the correct explanation of (A).

1. Assertion (A) : Degree of price elasticity is less than one in case of relatively inelastic demand.
   Reason (R) : Change in demand is less than the change in price.
2. Assertion (A) : A change in quantity demanded of one commodity due to a change in the price of other commodity is cross elasticity.
   Reason (R) : Changes in consumer income leads to a change in the quantity demanded.
3. Assertion (A) : Elasticity of demand explains that one variable is influenced by another variable.
   Reason (R) : The concept of elasticity of demand indicates the effect of price and changes in other factors on
4. Assertion (A) : Price elasticity is infinity in the case of a horizontal straight line demand curve.
   Reason (R) : Demand is more elastic in case of the long period as compared to a short period.
5. Assertion (A) : Luxury goods often have a high price elasticity of demand.
   Reason (R) : When the price of luxury goods increase, then their demand falls considerably as they are put to urgent uses only.

ANSWERS

1. (c)	2. (d)	3. (d)	4. (d)	5. (d)

2 MARK QUESTIONS

1. What does price elasticity of demand show?

Ans. The price elasticity of demand shows the responsiveness of demand of a commodity to the change in its price. Price elasticity of demand is measured by the ratio of percentage change in quantity demanded to a percentage change in price of the commodity.

$$ep = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ price}$$

2. What does income elasticity of demand show?

Ans. Income elasticity of demand measures the degree of responsiveness of quantity demanded of a commodity to a change in the income of the consumers. More precisely, income elasticity of demand may be defined as the ratio of the proportionate change in quantity demanded of a commodity to the proportionate change in the income of the consumers.

$$ey = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ income\ of\ the\ consumer}$$

3. What is cross elasticity of demand?

Ans. The degree of responsiveness of quantity demanded of one commodity to a change in the price of another commodity is called cross elasticity of demand.

$$exy = \frac{Percentage\ in\ quantity\ demanded\ of\ x\ commodity}{Percentage\ change\ in\ the\ price\ of\ Y\ commodity}$$

4. What is perfectly elastic demand?

Ans. When quantity demanded changes without any change in the price of the commodity and a little increase in the price brings the quantity demanded to zero. Demand curve in this case will be horizontal straight line parallel to X-axis and elasticity of demand will be infinite. It is an imaginary situation.

5. What is perfectly inelastic demand?

Ans. When there is no change in the quantity demanded of a commodity with the change in price of it is known as perfectly inelastic demand. The demand curve in this case will be a vertical straight line parallel to Y-axis and elasticity of demand will be zero. It can happen in the case of necessity without which the survival is difficult. But most of the economists consider it an imaginary situation.

6. What is elastic/highly elastic or more than unitary elastic demand?

Ans. When percentage change in quantity demanded is more than percentage change in price, it is known as highly elastic
demand. Demand curve in this case will be downward sloping & flatter and elasticity of demand will be > 1. e.g., Luxuries.

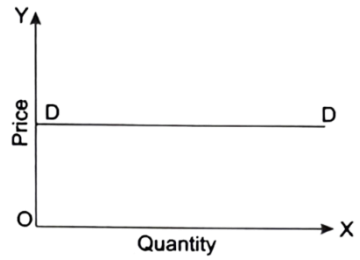
7. What is inelastic/highly inelastic or less than unitary elastic demand?

Ans. When percentage change in quantity demanded is less than percentage change in price, it is known as less elastic demand. Demand curve in this case will be downward sloping but steep and demand will be <1. It is an example of necessities.

8. What is unitary elastic demand?
Ans. When percentage change in quantity demanded is equal to percentage change in price, it is known as unitary elastic demand. Demand curve in this case will be rectangular hyperbola and elasticity of demand will be = 1.

9. Mention two factors affecting the price elasticity of demand.
Ans. The two factors which affect price elasticity of demand are as follows:
(i) Availability of substitutes: If a commodity has close substitutes, its demand is likely to be more elastic.
(i) Nature of the commodity: If the commodity is a necessity, its demand will be less elastic than a luxury item.

10. When is the elasticity of demand infinite?
Ans. When the change in demand is infinite without any change in price. Slightest rise in price may render the demand to zero.



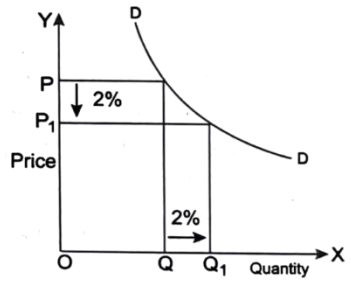
11. When is the demand for a commodity said to be unitary elastic?
Ans. When the proportional change in price brings equal proportional change in the quantity demanded.

12. State the percentage method of measuring elasticity of demand.
Ans. Under percentage method the price elasticity of demand is measured by using the formula

$$ep = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ price}$$

13. What are the degrees of elasticity of demand?
Ans. Degrees of price elasticity of demand are:
(i) Perfectly elastic demand
(ii) Perfectly inelastic demand
(iii) Highly elastic or relatively elastic or more than unitary elastic demand
(iv) Less elastic or inelastic or less than unitary elastic demand
(v) Unitary elastic demand

14. What is the shape of demand curve in case of unitary elastic demand?
Ans. In case of unitary elastic demand, the demand curve will be rectangular hyperbola (see graph).

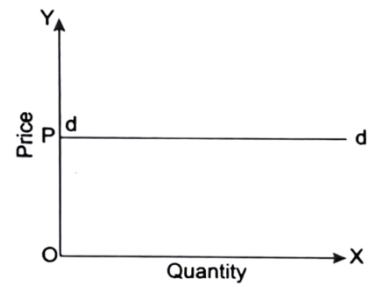


15. If two demand curves intersect which one will have higher elasticity?
Ans. If two demand curves intersect, then the flatter curve will be more elastic.

16. State the geometric/point method of calculating elasticity of demand.
Ans. Here price elasticity of demand at any point on the linear demand curve is

$$ep = \frac{Lower\ line\ segment\ of\ the\ demand\ curve}{Upper\ line\ segment\ of\ the\ demand\ curve}$$

17. Draw a perfectly elastic demand and tell about its shape.
Ans. Shape is horizontal straight line parallel to X-axis.



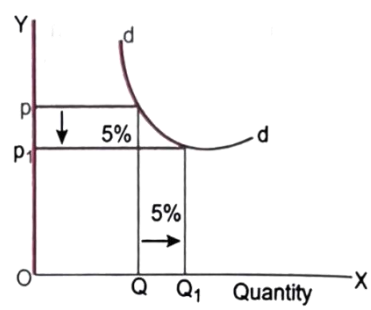
18. Give the formula to measure income elasticity of demand.
Ans. $ey = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ income\ of\ the\ consumer}$

$$ey = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

19. Give the formula to measure cross elasticity of demand.
Ans. $exy = \frac{Percentage\ change\ in\ the\ price\ of\ x\ commodity}{Percentage\ change\ in\ the\ price\ of\ Y\ commodity}$

$$exy = \frac{\Delta Qx}{\Delta Py} \times \frac{Py}{Qx}$$

20. Draw a demand curve known as rectangular hyperbola and give the reason for this shape of the demand curve.
Ans. When percentage change in quantity demanded is equal to percentage change in price, it is known as unitary elastic demand. Demand curve in this case will be rectangular hyperbola and elasticity of demand will be = 1. (Rectangular hyperbola is curve under which all rectangular areas will be the same)



21. Name the commodities which will have positive and negative cross elasticity of demand.
Ans. Substitute goods like tea and coffee will have positive cross elasticity of demand.
Complementary goods like pen and ink will have negative cross elasticity of demand.

22. Name the commodities which will have positive and negative income elasticity of demand.
Ans. Normal goods will have positive income elasticity, inferior goods will have negative income elasticity.

3-4 MARKS QUESTIONS

1. Give the three observations related to expenditure method of measuring elasticity of demand.

Ans. Under total expenditure method:

(i) Ep>1, if with fall in the price of the commodity total expenditure increases and with a rise in its price total expenditure decreases.

(ii) Ep <1, if rise or fall in the price of the commodity makes no change in total expenditure.

(iii) Ep=1, if with fall in the price of a commodity total expenditure decreases and with a rise in the price total expenditure increases.

2. Give three reasons for price elasticity of demand to be relatively inelastic.

Ans. When percentage change in quantity demanded is less than percentage change in price it is known as relatively less elastic or inelastic demand. It is due to following reasons:

(i) In case of necessary items the demand becomes price inelastic because consumption of these commodities can not be reduced to a large extent even if their prices rise to a substantial amount.

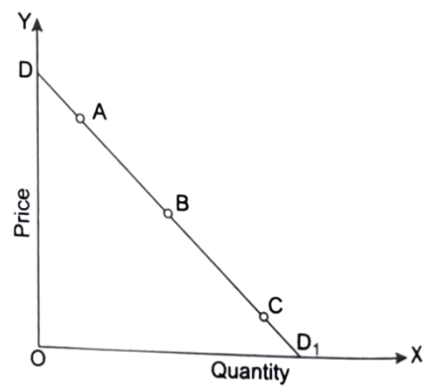
(ii) If the number of substitute items is relatively less in case of any commodity its demand will be relatively inelastic.

(iii) If the commodity has little number of alternative uses its demand will not rise to great extent even after a substantial fall in its price, so the demand will be less elastic.

3. Draw a straight line demand curve. Select any three points on it and compare the point elasticity at these three points.

Ans. On a straight line demand curve,

$$ep = \frac{Lower\ segment}{Upper\ segment}$$



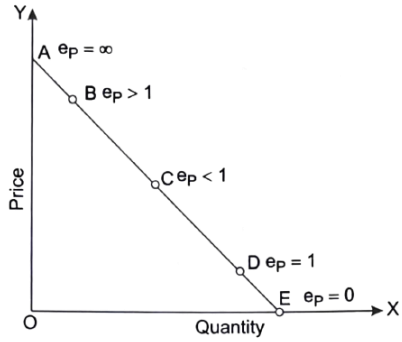
(i) At point A, the lower segment is bigger than the upper segment. Therefore, Ep>1.

(ii) At point B (the mid-point of the demand curve), lower segment is equal to the upper segment. Therefore, Ep = 1.

(iii) At point C, lower segment is smaller than the upper segment. Therefore, Ep <1.

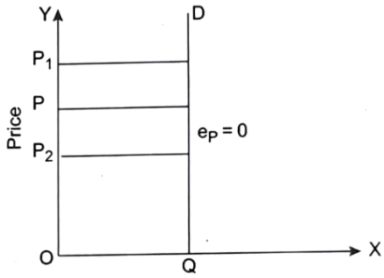
4. Draw a straight line demand curve touching both the axis. Divide it into two equal parts and mark the elasticity at all five points of it.

Ans.

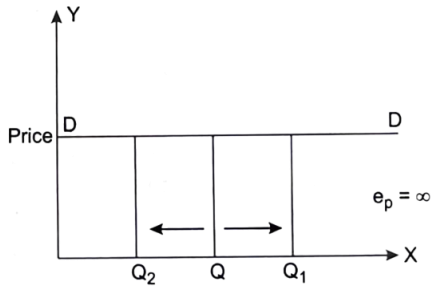


6-8 MARKS QUESTIONS

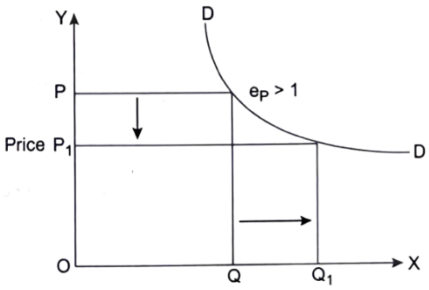
1. Explain the degrees of price elasticity of demand.



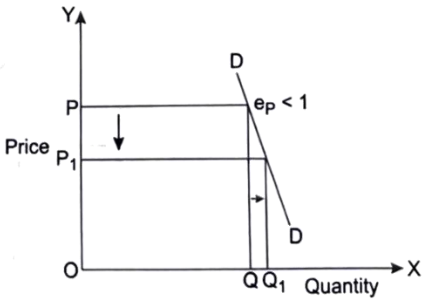
Ans. (i) Perfectly Inelastic Demand: When there is no change in the quantity demanded of a commodity with the change in price of, it is known as perfectly inelastic demand. The demand curve in this case will be a vertical straight line parallel to Y-axis and elasticity of demand will be zero. It can happen in case of necessities without which the survival is difficult. But most of the economists consider it an imaginary situation.



(ii) Perfectly Elastic Demand: When quantity demanded changes without any change in the price of the commodity and a little increase in the price brings the quantity demanded to zero. Demand curve in this case ako will be horizontal straight line parallel to X-axis and elasticity of demand will be infinite. It is an imaginary situation.

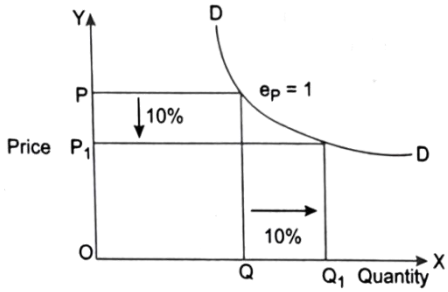


(iii) Highly Elastic or Relatively Elastic or More than Unitary Elastic: When percentage change in quantity demanded is more than percentage change in price, it is known as highly elastic demand. Demand curve in this case will be downward sloping and flatter and elasticity of demand will be > 1., e.g., Luxuries.



(iv) Less Elastic or Relatively Inelastic or Less than Unitary Elastic: When percentage change in quantity, demanded is less than percentage change in price, it is known as less elastic demand. Demand curve in this case will be downward sloping but steep and demand will be < 1. It is an example of necessities.

(v) Unitary Elastic Demand: When percentage change in quantity demanded is equal to percentage change in price, it is known as unitary elastic demand. Demand curve in this case will be rectangular hyperbola and elasticity of demand will be = 1 (Rectangular hyperbola is curve under which all rectangular areas will be the same).



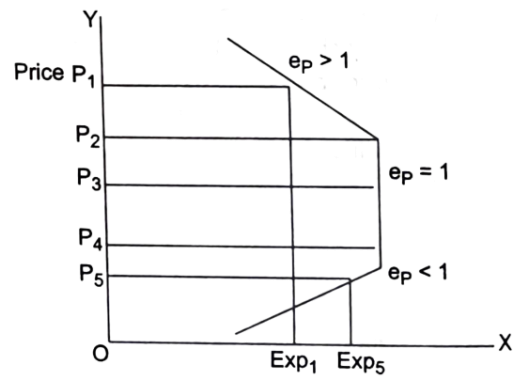
2. Explain total outlay/expenditure method with the help of numerical example.

Ans. Total Expenditure Method: It is given by professor Marshall. In this method, we measure the elasticity with the help of change in total expenditure due to change in price.

Table

Price	Demand	Total exp.	(PXD)
1	10	10	ep < 1
2	9	18	Total expenditure is increasing
3	8	24	ep < 1
4	7	28	
5	6	30	Total expenditure is constant
6	5	30	ep=1
7	4	28	Total expenditure is decreasing
8	3	24	ep > 1
9	2	18	
10	1	10	

There are three observations related to total expenditure method:
(i) When price and total expenditure both move in same direction. Either both are increasing or decreasing the elasticity in this case will be less than 1.
(ii) With the change in price the total expenditure does not change (remains Price P, constant), ep = 1.
(iii) When price and total expenditure move in opposite direction. When price increases expenditure decreases and vice-versa the elasticity of demand will be greater than 1.

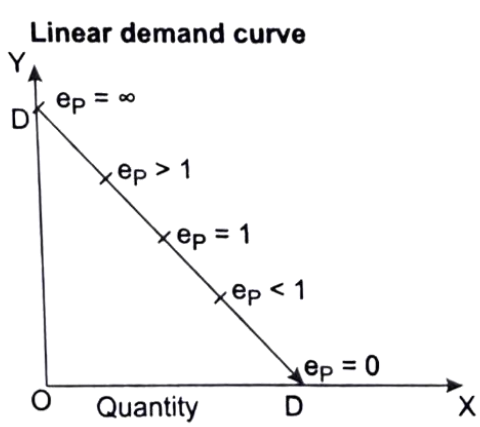


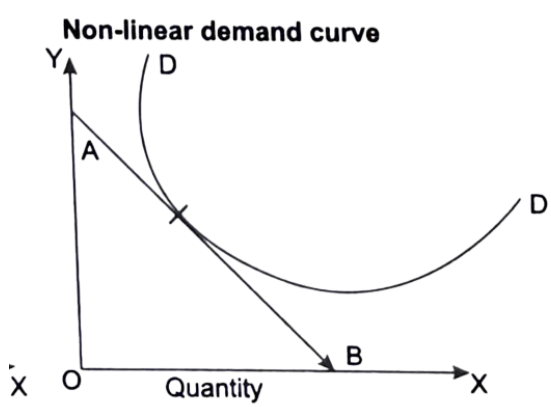
Total expenditure method does not give the exact magnitude of the elasticity it only give us general measure. By this method we only know whether the elasticity is equal to one, greater than one or less than one.

3. Explain point/geometric method to measure clasticity of demand.

Ans. Point Method/Geometric Method: In point method we measure the elasticity of demand of a particular point given on the demand curve. It is purely a geometric method,

$$ep = \frac{Lower\ line\ segment}{Upper\ line\ segment}$$





In order to calculate price elasticity of demand at any point on a curved demand curve, we have to draw a tangent to the demand curve through the chosen point and measure the elasticity of the tangent at this point as the ratio of the lower line segment to the upper line segment.

4. Explain percentage method to measure elasticity of demand.

Ans. In this method, price elasticity of demand is measured by the ratio of percentage change in quantity demanded to percentage change in price of the commodity. Thus,

$$ep = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ price} \times 100$$

$$ep = \frac{Change\ in\ quantity\ demanded}{Initial\ demand} \times 100$$

$$ep = \frac{Change\ in\ price}{Initial\ price} \times 100$$

$$= \frac{\frac{\Delta Q}{Q} \times 100}{\frac{\Delta P}{P} \times 100} = \frac{\Delta Q}{Q} \times \frac{P}{\Delta P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Percentage method does not depend on units of measurement of quantity demanded. It is unit free measurement because it uses percentage change in price and quantity demanded. While calculating price elasticity of demand by percentage method we ignore the negative signs. It means we should only take absolute values and not their signs.

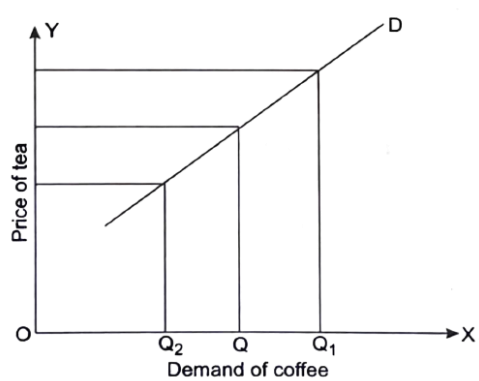
5. What do you mean by cross elasticity of demand? Explain the degrees of it.

Ans. The degree of responsiveness of quantity demanded of one commodity to the change in price of the other commodity is known as cross elasticity of demand.

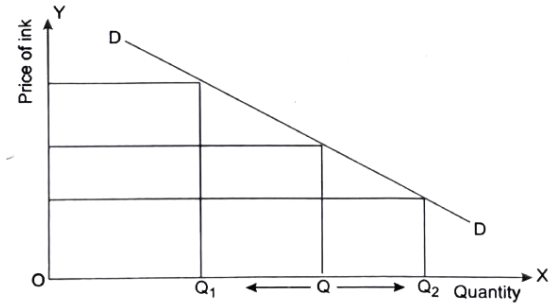
$$exy = \frac{Percentage\ change\ in\ quantity\ demanded\ of\ X\ commodity}{Percentage\ change\ in\ the\ price\ of\ Y\ commodity}$$

Types of Cross Elasticity of Demand:

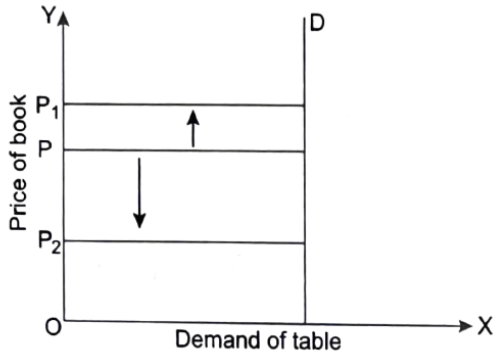
(i) Positive Cross Elasticity: When increase in price of one commodity leads to increase in quantity demanded of other commodity is known as positive cross elasticity of demand. It happens in the case of substitute goods.



(ii) Negative Cross Elasticity: When the increase in the price of one commodity leads to decrease in the demand of other commodity is known as negative cross elasticity of demand. It happens in the case of complementary goods. e.g., pen and ink, car and petrol.



(iii) Zero Cross Elasticity: When change in the price of one commodity does not bring any change in the quantity demanded of other commodity is called zero cross elasticity. It happens in the case of non-related goods, e.g., books and table.



6. Give any four factors affect elasticity of demand.

Ans. Price elasticity of demand is the degree of responsiveness of the quantity demanded of a good to a change in its price.

Factors Affecting Price Elasticity of Demand:

(i) **Number of Substitutes**: A commodity will have elastic demand if there are more no. of substitutes available, e.g., pepsi, coca-cola, and frooti. A commodity having less no. of substitutes, e.g., salt will have inelastic demand.

(ii) **Nature of the Commodity**: Generally, the demand for necessaries is inelastic (less elastic) and demand for luxuries is elastic (more elastic). This is so because certain goods which are essential to life will be demanded at any price, whereas goods meant for luxuries can be disposed off easily if they appear.

(iii) **Different Uses of Commodity**: If the commodity has different uses its demand will be elastic.

(iv) **Habits**: Commodities in habit of the consumers have less price elasticity.

(v) **Level of Income**: If consumer belongs to richer section then his demand will not be affected by change in price, hence demand will be inelastic.

QUESTIONS BASED ON DIFFERENTIATION

1. Differentiate between law of demand and elasticity of demand.

Ans.

S.No.	Basis	Law of Demand	Elasticity of Demand
1.	Meaning	Law of demand states, other things being equal, the inverse relation between price of a commodity and its quantity demanded.	Elasticity of demand is the rate of change in quantity demanded of a commodity as a result of change in its price.
2.	Change in demand	It reflects the direction of change in demand.	It tells us the magnitude of the change in demand as a result of change in price.

2. Distinguish between elastic and inelastic demand.

Ans. When the proportionate change in quantity demanded is more than the proportionate change in price, it is said to be elastic demand (ep > 1).

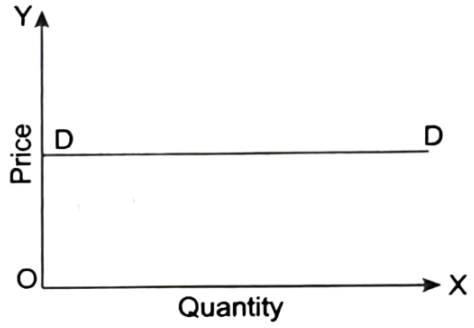
When the proportionate change in quantity demanded is less than the proportionate change in price is known as inelastic demand (ep < 1).

CASE BASED QUESTIONS

CASE 1

Answer the following questions on the basis of the adjoining graph :

1. Which type of elasticity of demand will have a demand curve given aside? Define it.
2. What will be the elasticity of demand in this case and why?
3. Which type of goods have such elasticity of demand?



Ans. 1. Perfectly elastic demand. In perfectly elastic demand, a small rise in price results in fall in demand to zero, while a small fall in price causes increase in demand to infinity. In such a case, the demand is perfectly elastic.

2. Elasticity of demand will be infinite. Ep = ∞. As price remains same and demand keeps on increasing.

3. Perfectly elastic demand is hypothetical. It can happen in case of any type of commodity.

CASE 2

Tea and coffee are substitutes of each other. They both can be used to satisfy same type of wants. With increase in the price of coffee, demand for tea will be increasing.

Answer the following questions based on the above statements:

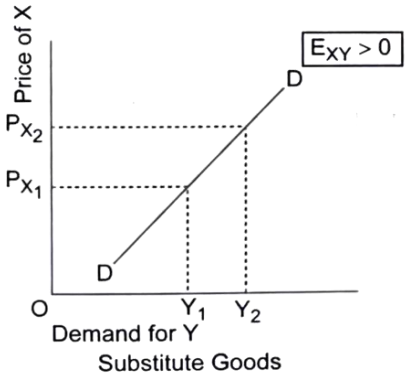
1. Define the elasticity of demand for the substitute goods.
2. Will a commodity with substitutes have elastic or inelastic demand and why?

3. Draw the demand curve showing elasticity between tea and coffee.

Ans. 1. Demand for substitute goods is positive elastic.

2. Elastic demand as if price of a commodity increases people will switch over to its cheaper substitutes.

3.



CASE 3

Salt has no close substitute and moreover a household has to spend a very small part of his entire budget. Therefore, even in spite of increase in price of salt, its, demand will not be affected. However, demand for a particular brand of toothpaste is comparatively more elastic as there are many brands of toothpaste available in the market, so the consumers can switch over to any other brand in case of hike in the price of a particular brand of toothpaste.

Answer the following questions based on the above paragraph:

1. Which factors according to above passage are responsible for price elasticity of demand?
2. Is the demand for a luxury item elastic and why?
3. Name some commodities, which have small number of substitutes.
4. What do you think about the price elasticity of demand for a particular brand of match box? Is it elastic or inelastic?

Ans. 1. Factors responsible for elasticity of demand:

(i) Proportion of income spent on the commodity.

(ii) Number of substitutes available of a commodity.

2. Elastic. If price of the luxury is high then the demand for it can be postponed.

3. Salt, Electricity etc.

4. It is inelastic as very small percentage of total income is spent on the matchboxes.

CASE 4

Elasticity of demand gets affected by many factors. Electricity has several uses of it. When price of it increases, people will demand for the most important uses of it only. On the other hand if its price falls, it will be demanded for all possible uses. There are many other goods also which show the same type of behaviour like milk, sugar, etc.

Answer the following questions based on the above paragraph:

1. What is elasticity of demand?
2. Explain any two factors affecting elasticity of demand.
3. What will be the elasticity of demand of electricity and why?

Ans. 1. Degree of responsiveness of quantity demanded of a commodity to the change in its determinants is known as elasticity of demand.
2. (i) Nature of the commodity
(ii) Time period
3. Demand for electricity will be highly elastic as it has several uses.

QUESTIONS WITH HIGH DIFFICULTY LEVEL

1. Why is the coefficient of price elasticity of demand is always negative?
Ans. The coefficient of price elasticity of demand is always negative because there is an inverse relationship between demand and price.

2. If two demand curves intersect which one will have higher elasticity?
Ans. If two demand curves intersect, then the flatter curve will be more elastic.

3. Draw a demand curve which has price elasticity equal to one every where.
Ans. When change in quantity demanded is equal to the change in price elasticity of demand will be equal to one at all the places of demand curve and shape of the demand curve will be rectangular hyperbola (see graph).

4. Show the effect of an increase in price on total expenditure depending on the values of price elasticity.
Ans. (i) If ep < 1 then with increase in the price total expenditure will also increase.
(ii) If ep> 1 then with increase in the price total expenditure will decrease.
(iii) If ep=1 then with increase in the price total expenditure will not change.

5. Is the demand for the following elastic, moderately elastic, highly elastic or completely inelastic? Give reasons for your answer.
(i) Demand for needles
(ii) Demand for petrol
(iii) Demand for cars
(iv) Demand for text books
(v) Demand for refrigerator
(vi) Demand for seasonal vegetables
(vii) Demand for milk.

Ans. (i) Demand for needles is completely inelastic. The reason being that the proportion of income spent on needle is very less. There will be no change in demand for needles with the change in its price.
(ii) Demand for petrol will be moderately elastic because if its price increases people will decrease the demand for it to some extent.
(iii) Demand for cars is elastic because car is a luxury item and its demand can be postponed if its price increases.
(iv) Demand for text books will be completely inelastic because they are necessities. Change in price will not have any effect on its demand.

(v) Demand for refrigerator will be completely elastic. The reason being that demand for it can be postponed with the increase in its price.
(vi) Demand for seasonal vegetables is less elastic because consumers would like to consume these vegetables even if their prices are high.
(iv) Demand for milk is elastic as it has several uses and when its prices increases it can be demanded only for some uses.
6. If demand becomes zero with a slight rise in the price, what would you call such a demand?
Ans. If demand becomes zero with a slight rise in the price, it is said to be perfectly elastic.
7. What do you call the curve under which area remains the same?
Ans. The curve under which area remains the same is known as rectangular hyperbola.
8. Will a downward sloping demand curve have the same or different price elasticity of demand at its different points?
Ans. A downward sloping demand curve has different price elasticity of demand at different points.
9. If two commodities are substitutes of each other, what will be the cross elasticity of demand for these?
Ans. The cross elasticity for these two goods will be positive.
10. If two commodities are complementary goods, what will be the cross elasticity of demand for these?
Ans. The cross elasticity of demand for these goods will be negative.
11. What does zero cross elasticity of demand between two goods imply? Give an example to explain.
Ans. Zero cross elasticity of demand between two goods implies that they are unrelated goods and the change in the price of one commodity does not affect the demand for another commodity. For example, books and TV.

NUMERICALS (SOLVED)

1. As the price of peanut packets increases by 5%, the number of peanut packets demanded falls by 8%. What is the elasticity of demand of peanut packets?

Solution. $ep = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ price} = 8/5 = 1.6$

ep = 1.6 Ans.

2. The quantity demanded of a commodity at a price of 8 per unit is 500 units. Its price falls to 6 and as a results its quantity demanded rises to 600 units. Calculate its price elasticity of demand.

Solution. Given values

ΔQ-100(600-500)Q=500 units

ΔP -2(6-8)Q_1 = 600 units

P=₹8

e_p-ΔQ/AP x P/Q

e_p=100/2 x 8/500-4/5=0.8

Ans.

3. At a price of 30 per unit, the quantity demanded of a commodity is 300 units. If price falls by 10%, its quantity demanded rises by 60 units. Calculate its price elasticity of demand.

Solution. Percentage change in price = 10

Percentage change in demand = $\frac{60}{300} \times 100 = 20$

$$ep = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ price} = 20/10 = 2$$

Ans.

4. At ₹ 5 per unit, consumer buys 40 units of a commodity and the price elasticity of demand is -2. How much will he buy if the price reduces to 4 per unit?

Solution. Given values; ep = -2, Q=40, P = ₹ 5, P_1= ₹ 4

ep=ΔQ/ΔP × P/Q

By putting the given values in the formula: 2=ΔQ/1 × 5/40 ⇒ 16=ΔQ

Thus, consumer will buy 56(40+16) units.

Ans.

5. Consumer buys 80 units of a good at a price of 5 per unit. Suppose price elasticity of demand is -2. At what price will he buy 64 units?

Solution. Given values; Q = 80 units, P= ₹ 5 per unit, e_p = -2, Q1=64 units, ΔQ=16 (80-64)

ep=ΔQ/ΔP x P/Q

$$-2 = \frac{80-64}{\Delta P} \times \frac{5}{8}$$

$$-2 = \frac{16}{\triangle P} \times \frac{5}{80}$$

$$\Delta P = 0.5$$

New price = 5 +0.5=5.5

Ans.

6. Price of a commodity falls from 4 per unit to 3 per unit. A result total expenditure on it rises from 200 to 300. Find out price elasticity of demand by method.

Solution.

Price(₹)	Total Expenditure (₹)	Quantity Demanded (Units)
200	4	200/4=50
300	3	300/3=100

$$Percentage\ change\ in\ price = \frac{3-4}{4} \times 100 = 25\%$$

$$Percentage\ change\ in\ quantity = \frac{100-50}{50} \times 100$$

$$ep = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ price} = 100/25 = 4$$

Ans.

7. The price elasticity of demand of good X is two times the price elasticity of demand of good Y. 8 percent fall in the price of Y results in 12 per cent rise in its demand. If the price of X rises by 5 per cent, calculate the percentage fall in its demand.

Solution. $ep(for\ good\ Y) = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ price} = 12/8 = 1.5$

$$ep(for\ good\ X) = 3(2 \times 1.5)$$

$$ep(for\ good\ X) = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ price}$$

$$3 = \frac{Percentage\ change\ in\ quantity\ demanded}{5}$$

15 = Percentage change in demand

Ans.

8. The price of a good is 20 per unit and total expenditure on it is 1000. When its price falls to 18 per unit, total expenditure rises by 8 per cent. Calculate price elasticity of demand using percentage method.

Solution.

Price(₹)	Total Expenditure (₹)	Quantity Demanded (Units)
20	1000	50(1000/20)
18	1080	60(1080/18)

New total expenditure= 1000+ 80 (1000 x 8/100)

Formula ep = $\Delta Q/\Delta P$X P/Q = 10/2 x 20/50 = 2

Ans.

9. A consumer buys 10 units of a commodity at a price of 10 per unit. He incurs an expenditure of 200 on buying 20 units. Calculate price elasticity of demand by the percentage method. Comment upon the shape of demand curve based on this information.

Solution.

Price(₹)	Total Expenditure (₹)	Quantity Demanded (Units)
10	10	100
10	20	200

Percentage method $e_p = \Delta Q/\Delta P$ x P/Q = 10/0 x 10/10 = 100/0 = ∞

The demand curve will be a horizontal line; that is, parallel to the X-axis. The value of ep remains infinity at all points of the curve.

Ans.

10. The percentage change in quantity demanded is 2 times the percentage change in price. At a price of 6 per unit, original quantity demanded was 20 units. If price rises by 10%, calculate the price elasticity of demand and new quantity demanded.

Solution. Percentage change in demand = 2 times of percentage change in price

Since, price rises by 10%, percentage fall in demand will be 20%.

$$ep = \frac{Percentage\ change\ in\ quantity\ demanded}{Percentage\ change\ in\ price} = 20/10 = 2$$

Ans.

Fall in demand(ΔQ) = 20% of 20 units = 20 x 20/100 = 4 units

New quantity demanded = Original quantity – ΔQ = 20 - 4 = 16 units

Ans.

NUMERICALS (UNSOLVED)

1. When price is ₹ 10 per unit, demand for a commodity is 100 units. As the price falls to ₹ 8 per unit, demand expands to 150 units. Calculate elasticity of demand.

[Ans. 2.5]

2. When price of sugar is ₹ 5 per kg, its demand is 50 kg but when the price rises to ₹ 10 per kg, its demand contracts to 40 kg. Calculate the elasticity of demand.
[Ans. 0.2]
3. Consider the demand for a good. At price ₹ 4, the demand for the goods is 25 units. Suppose price of the goods increases to ₹ 5 and as a result, the demand for the goods falls to 20 units. Calculate the price elasticity.
[Ans. 0.81]
4. The demand for a good falls to 240 units in response to rise in price by ₹ 2. If the original demand was 300 units at the price of ₹ 20, calculate price elasticity of demand.
[Ans. 2]
5. The market demand for a good at ₹ 4 per unit is 100 units. Due to increase in price, the market demand falls to 75 units. Find out the new price, if the price elasticity of demand is 1.
[Ans. New price =5]
6. When the price of good X is ₹ 5, the consumer buys 100 units of good X. At what price would he be willing to purchase 140 units of goods X? The price elasticity of demand for good X is ₹ 2.
[Ans. New price =74]
7. The demand for a good at ₹ 10 per unit is 40 units. Price falls by ₹ 5. If price elasticity of demand is (-) 3, calculate the new quantity demanded.
[Ans. New quantity = 100 units]
8. Price elasticity of demand for a product is 'unity'. A household buys 25 units of this product at the price of ₹5 per unit. If the price of product rises by 1. How much quantity of the product will the household buy?
[Ans. New quantity = 20 units]
9. When price of commodity falls by 80%, the quantity demanded of it increases by 100%. Find out its price elasticity of demand.
[Ans. ep> 1]
11. Calculate the coefficient of price elasticity of demand if 10% fall in the price of a commodity causes a 15% increase in its demand.
[Ans. ep > 1]
12. A 5% fall in the price of X leads to 10% rise in the demand for X. A 20% rise in the price of Y leads to 6% fall in the demand for Y. Calculate the price elasticities of demand of X and Y. Out of X and Y, which commodity is more elastic?
[Ans. ep=2]
13. Suppose the price elasticity of demand for a good is -0.2. If there is a 5% increase in the price of goods, by what percentage will the demand for the goods go down?
[Ans. Demand for the goods will fall by 1%]
14. A consumer buys 20 units of a good at ₹10 per unit. When its price falls by 10% its demand rises to 22 units. Find out the price elasticity of demand.
[Ans. ep=1]
15. The quantity demanded of a commodity at a price ₹ 8 per unit is 600 units. Its price falls by 25 per cent and the quantity demanded rises by 120 units. Calculate the price elasticity of demand. Is its demand elastic? Give reason for your answer.
[Ans. ep=0.8]

16. Calculate the price elasticity of demand for a commodity when its price increases by 25% and quantity demanded falls from 150 units to 120 units.
[Ans. ep=0.8]
17. The price of commodity is ₹15 per unit and its quantity demanded is 500 units. Its quantity demanded rises by 80 units as a result of fall in its price by 20 per cent. Calculate its price elasticity of demand. Is its demand inelastic?
[Ans. ep=0.8, demand is less elastic]
18. When the price of a commodity is ₹20 per unit, its quantity demanded is 800 units. When its price rises by ₹5 per unit, its quantity demanded falls by 20 per cent. Calculate its price elasticity of demand. Is its demand elastic? Give reasons for your answer.
[Ans. ep=0.8]
19. When price of a good falls from ₹5 to ₹3 per unit, its demand rises by 40%. Calculate its price elasticity of demand.
[Ans. ep=1]
20. Calculate price elasticity of demand by expenditure method:

Price ()	Total Expenditure ()
3	500
4	420

[Ans. ep> 1, because price and total expenditure move in two different directions.]
21. A consumer spends 80 on a commodity when its price is 1 per unit and spends 96 when its price is 2 per unit. What is the price elasticity of demand for the commodity?
[Ans. ep < 1, as price and total expenditure move in the same direction.]
22. If demand increased by 50% due to an increase in income by 75%. Calculate income elasticity of demand.
[Ans. 0.66]
23. Calculate income elasticity of demand from the following information:

Income of Vijay (per month)	Demand for Milk (per month)
3	12,000
4	14,000

24. A consumer decreases his demand from 100 units to 80 units of a commodity when his monthly income falls from ₹ 10,000 to ₹ 9,000. Calculate income elasticity of demand.

25. Calculate cross elasticity of demand with the help of the following information:

Price of Tea (X) (per kg)	Demand for Coffee (Y) (per kg)
50	2
70	3

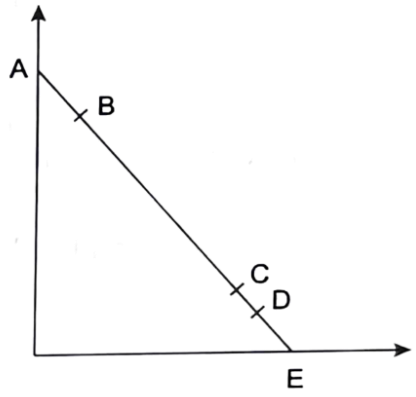
Also state whether cross elasticity of demand will be positive or negative.

26. Find the cross elasticity from the following data:

Price of Petrol (X) (per kg)	Demand for Cars (V) (per month)
30	100
70	120

WORKSHEET FOR SELF EVALUATION

1. The diagram shows AE is the demand curve of a commodity. On the basis of this diagram, state whether the following statements are true or false. Give reasons for your answer. **(4 Marks)**



(i) Demand at point B is price inelastic.
(i) Demand at point C is more price elastic than at point B.
(iii) Demand at point D is price inelastic.
(iv) Demand at point A is 0 and at point E is ∞,

2. What is the price elasticity of demand for following demand curve?
(i) Straight line demand curve parallel to X-axis;
(ii) Straight line demand curve parallel to Y-axis;
(iii) Mid-point of a straight line demand curve **(3 Marks)**

3. State with reasons, whether the following items will have elastic or inelastic demand: **(3 Marks)**

(i) Matchbox	(iii) Medicines	(v) Electricity
(ii) Coke	(iv) NCERT Textbook	(vi) Cigarettes

4. What happens to total expenditure on a commodity when its price falls and its demand is elastic? **(2 Marks)**

5. Draw (i) a demand curve which has unit elasticity everywhere, and (ii) another which has unit elasticity only at a single point. **(3 Marks)**

6. Draw demand curves with ep > 1, ep < 1. **(3 Marks)**

7. How is the elasticity of demand of a commodity affected by the following factors? **(3 Marks)**
(i) Existence of substitutes of a commodity
(ii) Nature of the commodity.

8. Show the types of income elasticity with the help of diagrams. **(6 Marks)**